

BOKARO STEEL PLANT — ILLUSTRATIVE SAMPLE

Standard Operating Procedure

SOP No: BSP-EL-005

Title: Electrical Isolation and Lockout-Tagout (LOTO)

Department: Electrical Maintenance

Revision: 02

Date: 27-Jun-2024

1. PURPOSE

To protect personnel from unexpected energisation or start-up of equipment during maintenance.

2. SCOPE

Applies to all electrical and electro-mechanical equipment: motors, conveyors, cranes, drives, panels and HT switchgear.

3. LOTO PROCEDURE

Step 1 — Inform the operator and stop the equipment through the normal stop button.

Step 2 — Obtain an Electrical Isolation Permit from the Electrical Shift In-charge.

Step 3 — Isolate at the source: rack out the breaker (HT) or switch off and remove fuses (LT).

Step 4 — Apply a personal padlock and a "DANGER — DO NOT OPERATE" tag. Every worker applies his or her own lock.

Step 5 — Verify zero energy: try to start the equipment locally (test for start) and test for voltage with an approved tester.

Step 6 — Discharge stored energy: capacitors, springs, hydraulic accumulators and gravity loads.

4. RULES

- One person, one lock, one key. Never lend your key.
- Locks may be removed only by the person who applied them.
- For HT equipment (above 650 V), earthing must be applied after isolation.

5. REMOVING LOTO

- Confirm all tools removed and guards refitted
- Confirm all persons clear
- Remove personal locks, return the permit, and inform the operator before restart

6. ELECTRIC SHOCK — FIRST RESPONSE

- Switch off the supply before touching the victim; if not possible, use a dry wooden stick
- Start CPR if the victim is not breathing and call Medical Centre: Ext. 2400